

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: B. Tech. Branch : Mechanical Engineering Semester : III

Subject Code & Name: Engineering Mathematics – III (BTBSC301)

Max Marks: 60

Date: / /2022

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		12
A)	Find $L\{f(t)\}$, where $f(t) = t^2 e^t \cos 2t$	Understand	6
B)	Find Laplace transform of periodic function $f(t) = E \sin at$ $0 < t < \frac{\pi}{a}$ and $f(t) = f(t + \frac{\pi}{a})$	Understand	6
C)	Using Laplace transform evaluate $\int_0^\infty e^{-2t} \frac{\sin^2 t}{t} dt$	Evaluation	6
Q.2	Solve Any Three of the following.		12
A)	Find $L^{-1}\left\{\log\left(\frac{s+a}{s+b}\right)\right\} = \dots\dots$	Application	4
B)	Find $L^{-1}\left\{\frac{s^2}{(s^2+3^2)(s^2+2^2)}\right\}$ using Convolution theorem.	Application	4
C)	Using Laplace transform solve $y'' + y = 6 \cos 3t$, $y(0) = 3$, $y'(0) = 1$	Application	4
D)	Solve $\frac{dx}{dt} = 2x - 3y$ and $\frac{dy}{dt} = y - 2x$, given $x(0) = 8$, $y(0) = 3$	Application	4
Q.3	Solve Any Two of the following.		12
A)	Find the Fourier integral representation of the function : $f(x) = 1, \quad x \leq 1$ $= 0, \quad x > 1$ Hence evaluate 1. $\int_0^\infty \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$, 2. $\int_0^\infty \frac{\sin \lambda}{\lambda} d\lambda$	Evaluation	6
B)	Find the Fourier Cosine integral representation of the function : $f(x) = x^2, \quad 0 < x < a$ $= 0, \quad x > a$	Understand	6
C)	Using the Fourier Sine Integral representation show that $e^{-2x} - e^{-3x} = \frac{10}{\pi} \int_0^\infty \frac{\lambda \sin \lambda x}{(\lambda^2+3^2)(\lambda^2+2^2)} d\lambda$	Understand	6
Q.4	Solve Any Three of the following.		12

	f from $f(xy + z^2, x + y + z) = 0$		
B)	Solve the partial differential equation $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$	Analysis	4
C)	Solve $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ subject to the condition $u(0, t) = 0 = (\pi, t)$ $u(x, 0) = 2 \sin 3x - 4 \sin 5x$	Application	4
D)	Solve $\frac{\partial^2 z}{\partial x^2} = 4 \frac{\partial^2 z}{\partial y^2}$ satisfying the condition $z = \frac{\partial z}{\partial y} = \sin x$, when $y = 0$	Application	4
Q. 5	Solve Any One of the following.		12
1.A	If $f(z)$ is analytic function and $f'(z) = 0$ then prove that $f(z)$ is a constant function.	Understand	6
1.B	Evaluate $\oint \frac{z+4}{z^2+2z+5} dz$ Over C, Where C is the circle $ z + 1 - i = 2$	Evaluation	6
	OR		
2.A	If $\phi = -x^3 + 3xy^2$ is flux function is an electrostatic field, find potential function and hence complex potential function.	Understand	6
2.B	Find the Residues of $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $ z = 2$	Understand	6
	*** End ***		

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Q.1]

A] Find $L\{f(t)\}$, where $f(t) = t^2 e^t \cos 2t$

Solⁿ: Let,

$$f(t) = \cos 2t$$

Taking L.T on both side

$$L\{f(t)\} = L\{\cos 2t\}$$

but,

$$L\{\cos at\} = \frac{s}{s^2 + a^2}$$

Here, $a = 2$

$$L\{\cos 2t\} = \frac{s}{s^2 + 2^2}$$

Next,

$$L\{t^2 \cos 2t\} = \frac{d}{ds} \left[\frac{d}{ds} \left(\frac{s}{s^2 + 2^2} \right) \right]$$

but,

$$\frac{d}{ds} \left(\frac{u}{v} \right) = \frac{vu' - uv'}{v^2}$$

$$u = s \quad u' = 1 \quad u'' = 0$$

$$v = s^2 + 4 \quad v' = 2s \quad v'' = 2$$

$$L\{t^2 \cos 2t\} = \frac{d}{ds} \left[\frac{(s^2 + 4)(1) - s(2s)}{(s^2 + 4)^2} \right]$$

$$L\{t^2 \cos 2t\} = \frac{d}{ds} \left[\frac{s^2 + 4 - 2s^2}{(s^2 + 4)^2} \right]$$

$$L \{ t^2 \cos 2t \} = \frac{d}{ds} \left[\frac{-s^2 + 4}{(s^2 + 4)^2} \right]$$

$$= - \frac{d}{ds} \left[\frac{s^2 - 4}{(s^2 + 4)^2} \right]$$

but,

$$\frac{d}{ds} \left(\frac{u}{v} \right) = \frac{v u' - u v'}{v^2}$$

$$u = s^2 - 4$$

$$u' = 2s$$

$$v = (s^2 + 4)^2$$

$$v' = 2(s^2 + 4)(2s)$$

$$L \{ t^2 \cos 2t \} = - \left[\frac{(s^2 + 4)^2 (2s) - (s^2 - 4) 2(s^2 + 4)(2s)}{(s^2 + 4)^4} \right]$$

$$L \{ t^2 \cos 2t \} = - (s^2 + 4) \left[\frac{(s^2 + 4)(2s) - (s^2 - 4)(4s)}{(s^2 + 4)^3} \right]$$

$$L \{ t^2 \cos 2t \} = - \left[\frac{(s^2 + 4)(2s) - (s^2 - 4)(4s)}{(s^2 + 4)^3} \right]$$

$$L \{ t^2 \cos 2t \} = - \left[\frac{2s^3 + 8s - (4s^3 - 16s)}{(s^2 + 4)^3} \right]$$

$$L \{ t^2 \cos 2t \} = - \left[\frac{2s^3 + 8s - 4s^3 + 16s}{(s^2 + 4)^3} \right]$$

$$L \{ t^2 \cos 2t \} = - \left[\frac{-2s^3 + 24s}{(s^2 + 4)^3} \right]$$

$$L \{ t^2 \cos 2t \} = \left[\frac{2s^3 - 24s}{(s^2 + 4)^3} \right]$$

$$L\{t^4 e^t \cos 2t\} = \left[\frac{2s^3 - 24s}{(s^2 + 4)^3} \right]_{s \rightarrow s-1}$$

$$L\{t^4 e^t \cos 2t\} = \left[\frac{2(s-1)^3 - 24(s-1)}{[(s-1)^2 + 4]^3} \right] //$$

6] Find Laplace transform of periodic function
 $f(t) = E \sin at \quad 0 < t < \frac{\pi}{a}$ and $f(t) = f\left(t + \frac{\pi}{a}\right)$

solⁿ: Given,

$$f(t) = E \sin at \quad 0 < t < \frac{\pi}{a} \text{ \& } f(t) = f\left(t + \frac{\pi}{a}\right)$$

This is the eqⁿ of periodic function with period
 $T = \frac{\pi}{a}$

we have,

$$L\{f(t)\} = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

$$= \frac{1}{1 - e^{-S(\frac{\pi}{a})}} \int_0^{\frac{\pi}{a}} e^{-st} \sin at dt$$

but,

$$\int_0^{\frac{\pi}{a}} e^{-st} \sin at dt$$

3] Using Laplace Transform evaluate $\int_0^{\infty} e^{-2t} \frac{\sin^2 t}{t} dt$

Solⁿ: Let,

$$\begin{aligned} f(t) &= \sin^2 t \\ &= \frac{1}{2} [1 - \cos 2t] \end{aligned}$$

Taking L.T on both side

$$L\{f(t)\} = \frac{1}{2} L\{1 - \cos 2t\}$$

$$= \frac{1}{2} L\{1\} - L\{\cos 2t\}$$

but,

$$L\{1\} = \frac{1}{s} \quad \& \quad L\{\cos at\} = \frac{s}{s^2 + a^2}$$

Here, $a = 2$

$$L\{f(t)\} = \frac{1}{2} \left\{ \frac{1}{s} - \frac{s}{s^2 + 2^2} \right\}$$

Next

$$L\left\{\frac{\sin^2 t}{t}\right\} = \frac{1}{2} \int_s^{\infty} \left[\frac{1}{s} - \frac{s}{s^2 + 2^2} \right] ds$$

but,

$$\int \frac{1}{s} ds = \log(s)$$

$$\int \frac{s}{s^2 + a^2} ds = \frac{1}{2} \log(s^2 + a^2)$$

Here, $a = 2$

$$\mathcal{L}\left\{\frac{\sin^2 t}{t}\right\} = \frac{1}{2} \left[\log(s) - \frac{1}{2} \log(s^2+4) \right]_s^\infty$$

$$= \frac{1}{2} \left[\log(s) - \log \sqrt{s^2+4} \right]_s^\infty$$

$$= \frac{1}{2} \left\{ \left[\log(\infty) - \log \sqrt{\infty^2+4} \right] - \left[\log(s) - \log \sqrt{s^2+4} \right] \right\}$$

$$= -\frac{1}{2} \left[\log(s) - \log \sqrt{s^2+4} \right]$$

$$= -\frac{1}{2} \left[\log \frac{s}{\sqrt{s^2+4}} \right]$$

Now,

$$\int_0^\infty e^{-st} \left\{ \frac{f(t)}{t} \right\} dt = \int_0^\infty e^{-2t} \left\{ \frac{\sin^2 t}{t} \right\} dt$$

$$\int_0^\infty e^{-2t} \frac{\sin^2 t}{t} dt = -\frac{1}{2} \left[\log \frac{s}{\sqrt{s^2+4}} \right] \quad \text{Put } s=2$$

$$= -\frac{1}{2} \left[\log \frac{2}{\sqrt{2^2+4}} \right]$$

$$= -\frac{1}{2} \left[\log \frac{2}{2\sqrt{2}} \right]$$

$$\int_0^\infty e^{-2t} \frac{\sin^2 t}{t} dt = -\frac{1}{2} \log\left(-\frac{1}{\sqrt{2}}\right), //$$

Q.2]

A] Find $L^{-1} \left\{ \log \left(\frac{s+a}{s+b} \right) \right\} = \dots\dots\dots$

Soln: Let,

$$f(s) = \log(s+a) - \log(s+b)$$

diff w.r.t s

$$\frac{d}{ds} f(s) = \frac{d}{ds} \log(s+a) - \frac{d}{ds} \log(s+b)$$

$$\frac{d}{ds} f(s) = \frac{1}{s+a} - \frac{1}{s+b}$$

Taking I.L.T on B.S

$$L^{-1} \left\{ \frac{d}{ds} f(s) \right\} = L^{-1} \left\{ \frac{1}{s+a} \right\} - L^{-1} \left\{ \frac{1}{s+b} \right\}$$

$$\Rightarrow -t f(t) = e^{-at} - e^{-bt}$$

$$f(t) = \frac{e^{-at} - e^{-bt}}{-t}$$

$$f(t) = \frac{e^{-bt} - e^{-at}}{t}$$

$$f(t) = \frac{e^{-bt} - e^{-at}}{t}$$

Q] Find $L^{-1} \left\{ \frac{s^2}{(s^2+3^2)(s^2+2^2)} \right\}$ using convolution theorem.

Solⁿ: Let,

$$f(s) = \frac{s^2}{(s^2+3^2)(s^2+2^2)} = \frac{s}{(s^2+3^2)} \times \frac{s}{(s^2+2^2)}$$

where,

$$f_1(s) = \frac{s}{(s^2+3^2)} \xrightarrow{\text{Taking I.L.T}} L\{f_1(s)\} = L\left\{\frac{s}{(s^2+3^2)}\right\}$$

$$f_1(t) = \cos 3t$$

$$f_2(s) = \frac{s}{(s^2+2^2)} \xrightarrow{\text{Taking I.L.T}} L\{f_2(s)\} = L\left\{\frac{s}{(s^2+2^2)}\right\}$$

$$f_2(t) = \cos 2t$$

$$f_1(t) = \cos 3t \Rightarrow f_1(u) = \cos 3u$$

$$f_2(t) = \cos 2t \Rightarrow f_2(t-u) = \cos 2(t-u) \\ = \cos(2t-2u)$$

we know,

$$L^{-1}\{f_1(s) \cdot f_2(s)\} = \int_0^t f_1(u) f_2(t-u) du$$

$$= \int_0^t \cos 3u \cdot \cos(2t-2u) du$$

but,

$$\cos A \cdot \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$\cos 3u \cdot \cos(2t-2u) = \frac{1}{2} [\cos(3u+2t-2u) + \cos(3u-2t+2u)]$$

$$L^{-1}\{f(s)\} = \frac{1}{2} \int_0^t \cos(2t+u) + \cos(5u-2t) \, du$$

$$= \frac{1}{2} \left[\frac{\sin(2t+u)}{1} + \frac{\sin(5u-2t)}{5} \right]_0^t$$

$$L^{-1}\{f(s)\} = \frac{1}{2} \left\{ \left[\sin(2t+t) + \sin(5t-2t) \right] - \left[\sin(2t) + \frac{\sin(-2t)}{5} \right] \right\}$$

$$L^{-1}\{f(s)\} = \frac{1}{2} \left\{ \sin(3t) + \sin(3t) - \sin(2t) + \frac{\sin 2t}{5} \right\}$$

$$L^{-1}\{f(s)\} = \frac{1}{2} \left\{ 2\sin(3t) - \sin 2t + \frac{\sin 2t}{5} \right\}$$

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c] Using Laplace transform solve $y'' + y = 6 \cos 3t$.
 $y(0) = 3, \quad y'(0) = 1$

soln:

Let,

$$y'' + y = 6 \cos 3t$$

Taking L.T on both side

$$L\{y''(t)\} + L\{y(t)\} = 6L\{\cos 3t\}$$

$$s^2 y(s) - sy(0) - y'(0) + y(s) = 6 \left[\frac{s}{s^2 + 3^2} \right]$$

$$s^2 y(s) - s(3) - (1) + y(s) = 6 \left[\frac{s}{s^2 + 3^2} \right]$$

$$s^2 y(s) - 3s - 1 + y(s) = 6 \left[\frac{s}{s^2 + 9} \right]$$

$$(s^2 + 1) y(s) - 3s - 1 = 6 \left[\frac{s}{s^2 + 9} \right]$$

$$(s^2 + 1) y(s) = \frac{6s}{s^2 + 9} + 3s + 1$$

$$y(s) = \frac{6s}{(s^2 + 9)(s^2 + 1)} + \frac{3s + 1}{(s^2 + 1)} \quad \dots \textcircled{1}$$

$$\frac{6s}{(s^2+9)(s^2+1)} = \frac{As+B}{s^2+9} + \frac{Cs+D}{s^2+1} \quad \dots (2)$$

$$6s = (As+B)(s^2+1) + (Cs+D)(s^2+9) \quad \dots (3)$$

$$6s = As^3 + As + Bs^2 + B + Cs^3 + 9Cs + Ds^2 + 9D$$

$$6s = (A+C)s^3 + (B+D)s^2 + (A+9C)s + B+9D$$

comparing coefficient

$$A+C=0 \quad \dots (4)$$

$$B+D=0 \quad \dots (5)$$

$$A+9C=6 \quad \dots (6)$$

$$B+9D=0 \quad \dots (7)$$

eqn (4) - eqn (6)

$$A+C=0$$

$$A+9C=6$$

$$-8C = -6$$

$$8C = 6$$

$$C = \frac{6^3}{84}$$

$$C = \frac{3}{4}$$

from eqn (4)

$$A + \frac{3}{4} = 0$$

$$A = -\frac{3}{4}$$

$$\text{eqn (5)} - \text{eqn (7)}$$

$$B + D = 0$$

$$B + 9D = 0$$

$$-8D = 0$$

$$D = 0$$

from eqn (5)

$$B + 0 = 0$$

$$B = 0$$

from eqn (2)

$$\frac{6s}{(s^2+9)(s^2+1)} = \frac{-\frac{3}{4}s + 0}{s^2+9} + \frac{\frac{3}{4}s + 0}{s^2+1}$$

$$\frac{s}{(s^2+9)(s^2+1)} = -\frac{3}{4} \left[\frac{s}{s^2+9} \right] + \frac{3}{4} \left[\frac{s}{s^2+1} \right]$$

from eqn (1)

$$y(s) = -\frac{3}{4} \left[\frac{s}{s^2+9} \right] + \frac{3}{4} \left[\frac{s}{s^2+1} \right] + \frac{3s+1}{s^2+1}$$

$$y(s) = -\frac{3}{4} \left[\frac{s}{s^2+9} \right] + \frac{3}{4} \left[\frac{s}{s^2+1} \right] + 3 \left[\frac{s}{s^2+1} \right] + \frac{1}{s^2+1}$$

Taking I.L.T on B.S

$$L^{-1}\{f(s)\} = -\frac{3}{4} L^{-1}\left\{\frac{s}{s^2+9}\right\} + \frac{3}{4} L^{-1}\left\{\frac{s}{s^2+1}\right\} \\ + 3 L^{-1}\left\{\frac{s}{s^2+1}\right\} + L^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$L^{-1}\{f(s)\} = -\frac{3}{4} \cos 3t + \frac{3}{4} \cos t + 3 \cos t + \sin t$$

Q] Solve $\frac{dx}{dt} = 2x - 3$ and $\frac{dy}{dt} = y - 2x$ given $x(0) = 8$,
 $y(0) = 3$

Q.3] A] Find the fourier integral representation of the function $f(x) = \begin{cases} 1 & |x| \leq 1 \\ 0 & |x| > 1 \end{cases}$ Hence evaluate

1. $\int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$ 2. $\int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda$

Solⁿ: we have by using cosine integral Representation

$$f(x) = \int_0^{\infty} A(\lambda) \cos \lambda x d\lambda \quad \text{--- (1)}$$

where,

$$A(\lambda) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \lambda x dx \quad \text{--- (2)}$$

from eqⁿ (2)

$$A(\lambda) = \frac{1}{\pi} \int_{-1}^1 (1) \cos \lambda x dx$$

$$= \frac{1}{\pi} \left[\frac{\sin \lambda x}{\lambda} \right]_{-1}^1$$

$$= \frac{1}{\pi} \left[\frac{\sin \lambda}{\lambda} + \frac{\sin \lambda}{\lambda} \right]$$

$$A(\lambda) = \frac{2}{\pi} \left[\frac{\sin \lambda}{\lambda} \right]$$

from eqⁿ (D)

$$f(x) = \frac{e}{\pi} \int_0^{\infty} \frac{\sin \lambda}{\lambda} \cos \lambda x \, d\lambda$$

$$\frac{\pi}{2} f(x) = \int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} \, d\lambda$$

$$\int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} \, d\lambda = \frac{\pi}{2}$$

Put $x=0$

$$\int_0^{\infty} \frac{\sin \lambda}{\lambda} \cos(0) \, d\lambda = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\sin \lambda}{\lambda} \, d\lambda = \frac{\pi}{2}$$

Q.3]

Q] Find the Fourier Cosine integral representation of the function:

$$f(x) = \begin{cases} x^2 & 0 < x < a \\ 0 & x > a \end{cases}$$

Solⁿ: we have by using Cosine Integral Representation

$$f(x) = \int_0^{\infty} A(\lambda) \cos \lambda x \, d\lambda \quad \dots \textcircled{1}$$

$$A(\lambda) = \frac{2}{\pi} \int_0^{\infty} f(x) \cos \lambda x \, dx \quad \dots \textcircled{2}$$

Here,

$$f(x) = x^2$$

from eqⁿ (2)

$$A(\lambda) = \frac{2}{\pi} \int_0^a x^2 \cos \lambda x \, dx$$

but,

$$\int u v \, dx = u v_1 - u' v_2 + u'' v_3 - u''' v_4 \dots \dots$$

Here,

$$u = x^2 \quad u' = 2x \quad u'' = 2$$

$$v = \cos \lambda x \quad v_1 = \frac{\sin \lambda x}{\lambda} \quad v_2 = -\frac{\cos \lambda x}{\lambda^2} \quad v_3 = -\frac{\sin \lambda x}{\lambda^3}$$

$$A(\lambda) = \frac{2}{\pi} \left\{ x^2 \frac{\sin \lambda x}{\lambda} - 2x \frac{-\cos \lambda x}{\lambda^2} + 2 \frac{-\sin \lambda x}{\lambda^3} \right\}_0^a$$

$$A(x) = \frac{2}{\pi} \left\{ x^2 \frac{\sin \lambda x}{1} + 2x \frac{\cos \lambda x}{\lambda^2} - 2 \frac{\sin \lambda x}{\lambda^3} \right\}_0^{\infty}$$

$$= \frac{2}{\pi} \left\{ \left[a^2 \frac{\sin \lambda a}{1} + 2a \frac{\sin \lambda a}{\lambda^2} - 2 \frac{\sin \lambda a}{\lambda^3} \right] - \right.$$

$$\left. \left[0 + 0 - 0 \right] \right\}$$

$$A(x) = \frac{2}{\pi} \left[a^2 \frac{\sin \lambda a}{1} + 2a \frac{\sin \lambda a}{\lambda^2} - 2 \frac{\sin \lambda a}{\lambda^3} \right]$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \left[a^2 \frac{\sin \lambda a}{1} + 2a \frac{\sin \lambda a}{\lambda^2} - 2 \frac{\sin \lambda a}{\lambda^3} \right] \cos \lambda x \, d\lambda$$

c] Using the Fourier sine integral representation show that

$$e^{-2x} - e^{-3x} = \frac{20}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x}{(\lambda^2 + 2^2)(\lambda^2 + 3^2)} d\lambda$$

Solⁿ: By using Fourier sine integral representation

$$f(x) = \int_0^{\infty} B(\lambda) \sin \lambda x d\lambda \quad \text{--- (1)}$$

where,

$$B(\lambda) = \frac{2}{\pi} \int_0^{\infty} f(x) \sin \lambda x dx \quad \text{--- (2)}$$

Here,

$$f(x) = e^{-2x} - e^{-3x}$$

from eqⁿ (2)

$$B(\lambda) = \frac{2}{\pi} \int_0^{\infty} e^{-2x} \sin \lambda x dx - \int_0^{\infty} e^{-3x} \sin \lambda x dx$$

but,

$$\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

here,

$$a = 2$$

$$a = 3$$

$$b = 1$$

$$b = 1$$

$$B(\lambda) = \frac{2}{\pi} \left[\frac{1}{(\lambda^2 + 2^2)} - \frac{1}{(\lambda^2 + 3^2)} \right]$$

$$B(\lambda) = \frac{2}{\pi} \left[\frac{\lambda(\lambda^2+3^2) - \lambda(\lambda^2+2^2)}{(\lambda^2+2^2)(\lambda^2+3^2)} \right]$$

$$= \frac{2}{\pi} \left[\frac{\lambda^3+9\lambda - \lambda^3-4\lambda}{(\lambda^2+2^2)(\lambda^2+3^2)} \right]$$

$$= \frac{2}{\pi} \left[\frac{5\lambda}{(\lambda^2+2^2)(\lambda^2+3^2)} \right]$$

$$B(\lambda) = \frac{10}{\pi} \left[\frac{1}{(\lambda^2+2^2)(\lambda^2+3^2)} \right]$$

from eqn (1)

$$f(x) = \int_0^{\infty} \frac{10}{\pi} \left[\frac{1}{(\lambda^2+2^2)(\lambda^2+3^2)} \right] \sin \lambda x \, d\lambda$$

$$e^{-2x} - e^{-3x} = \frac{10}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x}{(\lambda^2+2^2)(\lambda^2+3^2)} \, d\lambda$$

Q.4

A] From the partial differential equation by eliminating the arbitrary function from $(x+y+z^2, x+y+z)=0$

solⁿ: Let,

$$u = x+y+z^2$$

$$v = x+y+z$$

$$f(u, v) = 0 \quad \dots (1)$$

solⁿ of eqⁿ (1)

$$Pp + Qq = R \quad \dots (2)$$

diff u & v w.r.t x, y, z

$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial v}{\partial x} = 1$$

$$\frac{\partial u}{\partial y} = 1$$

$$\frac{\partial v}{\partial y} = 1$$

$$\frac{\partial u}{\partial z} = 2z$$

$$\frac{\partial v}{\partial z} = 1$$

$$P = \frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial y}$$

$$= 1(1) - 2z(1)$$

$$P = 1 - 2z$$

$$Q = \frac{\partial u}{\partial z} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z}$$

$$= 22(1) - y(1)$$

$$Q = 22 - y$$

$$R = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}$$

$$= y(1) - x(1)$$

$$R = y - x$$

from eqⁿ (2)

$$(x - 22)_p + (22 - y)_q = (y - x)$$

Q.5]

1. A] If $f(z)$ is analytic function and $f'(z)=0$ then P.T $f(z)$ is constant function.

Solⁿ: Given,

$f(z)$ is an analytic

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \dots (1)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \dots (2)$$

This C-R equation are satisfied

now,

$$f(z) = u(x,y) + i v(x,y) \quad \dots (3)$$

diff eqⁿ (3) w.r.t x

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = 0 \quad \dots \text{given}$$

i.e

$$\frac{\partial u}{\partial x} = 0 \quad \& \quad \frac{\partial v}{\partial x} = 0$$

from eqⁿ (3) & (3)

$$0 = \frac{\partial v}{\partial y} \quad \dots (4)$$

$$\frac{\partial u}{\partial y} = 0 \quad \dots (5)$$

on Integrating eqn (3)

$$f(z) = \int u(x,y) dx + i \int v(x,y) dx$$

$$f(z) = c + ic$$

There fore $f(z)$ is a constant function.

1.8] Evaluate $\oint \frac{z+4}{z^2+2z+5} dz$ over C where C is the circle $|z+1-i|=2$

Soln: we have Cauchy Integral formula

$$\oint_C \frac{f(z)}{(z-a)^{n+1}} dz = \frac{2\pi i}{n!} f^n(a) \quad \dots (1)$$

Given,

$$\oint_C \frac{z+4}{z^2+2z+5} dz \quad \dots (2)$$

Denominator

$$z^2 + 2z + 5$$

$$z = -1+2i, \quad z = -1-2i$$

from eqn (2)

$$\int_C \frac{z+4}{[z+(1-2i)][z+(1+2i)]} dz$$

Here,

$$f(z) = z+4$$

$$\alpha = -1+2i \quad n=0$$

$$\alpha = -1-2i \quad n=0$$

$$f^n(\alpha) = f(z)$$

$$f(-1+2i) = -1+2i+4 \\ = 3+2i$$

also,

$$f(-1-2i) = -1-2i+4 \\ = 3-2i$$

By using partial fraction method

$$\frac{1}{[z+(1-2i)][z+(1+2i)]} = \frac{A}{z+(1-2i)} + \frac{B}{z+(1+2i)} \quad \dots (3)$$

$$1 = A[z+(1+2i)] + B[z+(1-2i)] \quad \dots (4)$$

Put,

$$z+1+2i=0 \Rightarrow z=-1-2i \text{ in eqn (4)}$$

$$1 = B[-1-2i+1+2i]$$

$$1 = B[-4i]$$

$$B = -\frac{1}{4i}$$

Now, put $z+1-2i=0 \Rightarrow z=-1+2i$ in eqn (4)

$$1 = A[-1+2i+1+2i]$$

$$1 = A[4i]$$

$$A = \frac{1}{4i}$$

from eqn (3)

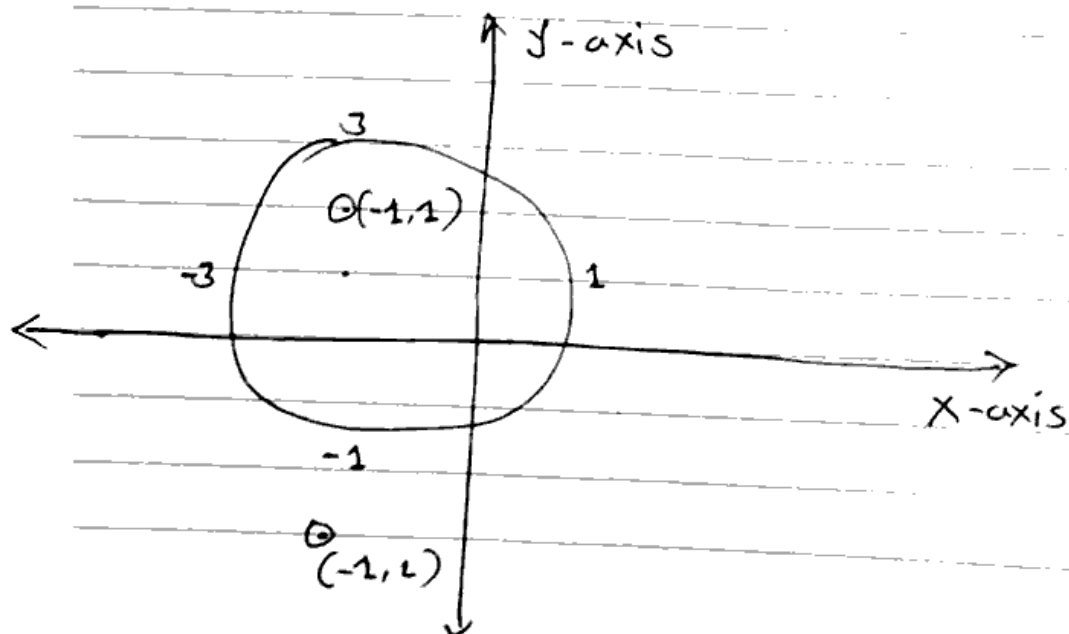
$$\frac{1}{[z+(1-2i)][z+(1+2i)]} = \frac{1}{4i} \left[\frac{1}{[z+(1-2i)]} \right] - \frac{1}{4i} \left[\frac{1}{[z+(1+2i)]} \right]$$

$$\int_C \frac{z+4}{[z+(1-2i)][z+(1+2i)]} dz = \frac{1}{4i} \int_C \frac{z+4}{[z+(1-2i)]} dz - \frac{1}{4i} \int_C \frac{z+4}{[z+(1+2i)]} dz$$

Now,

$$|z+1-i|=2$$

This is the eqn of circle with center $(-1, 1)$ & radius 2



Here,

$$a = -1 + 2i = (-1, 2)$$

Inside the circle

$$a = -1 - 2i = (-1, -2)$$

outside the circle

Hence

$$\oint_C \frac{z+4}{[z+(1-2i)][z+(1+2i)]} dz = \frac{1}{4i} \left[2\pi i f(a) \right] - \frac{1}{4i} [0]$$

$$= \frac{\pi}{2} [f(a)]$$

$$= \frac{\pi}{2} (3+2i)$$

$$\oint_C \frac{z+4}{z^2+2z+5} dz = \frac{\pi}{2} (3+2i)$$

2. A] If $\psi = -x^3 + 3xy^2$ is flux function on electrostatic field, find potential function & hence complex potential function.

solⁿ: Let,

$$f(z) = u + iv$$

$$\text{put } \phi = u \quad \& \quad \psi = v$$

Here,

$$v = -x^3 + 3xy^2$$

diff partially w.r.t x & y

$$\frac{\partial V}{\partial x} = -3x^2 + 3y^2$$

$$\frac{\partial V}{\partial y} = 6xy$$

By using Milne Thomson method

$$f(z) = \int \psi_1(z, 0) dz + i \int \psi_2(z, 0) dz + C \quad \dots (1)$$

where,

$$\psi_1(z, 0) = v_y$$

$$\psi_2(z, 0) = v_x$$

Now,

$$\begin{aligned} \psi_1(z, 0) &= 6(z)(0) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \psi_2(z, 0) &= -3(\cancel{z})^2 + 3(0)^2 \\ &= -3z^2 \end{aligned}$$

from eqn (1)

$$\begin{aligned} f(z) &= \int 0 dz + i \int -3z^2 dz + C \\ &= -3i \int z^2 dz + C \end{aligned}$$

$$f(z) = -8i \left[\frac{z^3}{3} \right] + C$$

$$= -i [z^3]$$

$$f(z) = -i (x+iy)^3 + C$$

$$= -i [x^3 + 3ix^2y + 3i^2xy^2 + i^3y^3] + C$$

$$= -i [x^3 + 3ix^2y - 3xy^2 - iy^3] + C$$

$$= -ix^3 - i^2 3x^2y + 3ixy^2 + i^2 y^3 + C$$

$$= -ix^3 + 3x^2y + 3ixy^2 - y^3 + C$$

$$f(z) = [3x^2y - y^3] + i[-x^3 + 3xy^2] + C$$

$$\phi = 3x^2y - y^3$$

2.6] Find the Residue of $f(z) = \frac{\sin z}{z \cos z}$ at its

poles inside the circle $|z| = 2$

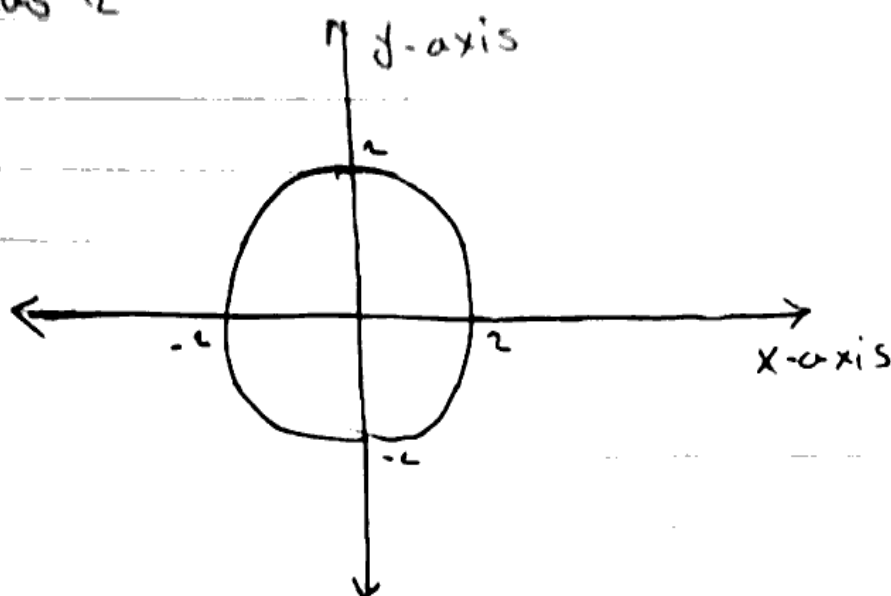
Soln: Let,

$$f(z) = \frac{\sin z}{z \cos z}$$

$$z = 0, \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$$

Here,

$|z|=2$ This is the eqⁿ of circle with center 0 & radius 2



$z_0 = 0$ Inside the circle

$z_0 = \pm \frac{\pi}{2} = \pm \frac{3.14}{2} = \pm 1.57$ Inside the circle

$z_0 = \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots$ outside the circle

Residue at $z_0 = 0$

we have,

$$\text{Res } z = z_0^{f(z)} = \lim_{z \rightarrow z_0} (z - z_0) f(z) \quad \dots \text{--- (1)}$$

$$\text{Res } z = 0^{f(z)} = \lim_{z \rightarrow 0} (z - 0) \frac{\sin z}{z \cos z}$$

$$\begin{aligned} \text{Res } z = 0 f(z) &= \frac{\sin(0)}{\cos(0)} \\ &= \frac{0}{1} \end{aligned}$$

$$\boxed{\text{Res } z = 0 f(z) = 0}$$

Residue at $z_0 = \frac{\pi}{2}$

from eqn (1)

$$\text{Res } z = \frac{\pi}{2} f(z) = \lim_{z \rightarrow \frac{\pi}{2}} \left(z - \frac{\pi}{2} \right) \cdot \frac{\sin z}{z \cos z} \left[\frac{0}{0} \text{ form} \right]$$

$$= \lim_{z \rightarrow \frac{\pi}{2}} \left[\frac{(z - \frac{\pi}{2}) \cos z + \sin z (1)}{z (-\sin z) + \cos z (1)} \right]$$

$$= \frac{\sin \frac{\pi}{2}}{\frac{\pi}{2} (-\sin \frac{\pi}{2}) + \cos \frac{\pi}{2}}$$

$$= \frac{1}{-\frac{\pi}{2}}$$

$$\boxed{\text{Res } z = \frac{\pi}{2} f(z) = -\frac{2}{\pi}}$$

Residue at $z_0 = -\frac{\pi}{2}$

from eqⁿ (1)

$$\text{Res } z = -\frac{\pi}{2} f(z) = \lim_{z \rightarrow -\pi/2} \left(z + \frac{\pi}{2} \right) \frac{\sin z}{z \cos z}$$

$$= \lim_{z \rightarrow -\pi/2} \left[\frac{(z + \pi/2) \cos z + \sin z (1)}{z (-\sin z) + \cos z (1)} \right]$$

$$= \frac{-\sin \pi/2}{-\pi/2 (\sin \pi/2) + \cos \pi/2}$$

$$= \frac{1}{\pi/2}$$

$$\boxed{\text{Res } z = -\frac{\pi}{2} f(z) = \frac{2}{\pi}}$$

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